Cardiac Field Artifact Removal (CFAR): A Large-Cohort Study Cleaning the Heart's Noise from Brain Signals

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Abstract

The heart's electrical field reaches the scalp as a cardiac field artifact (CFA) in every EEG recording, adding noise to any analysis and confounding heartbeat-evoked potentials (HEPs). CFA has been studied only in small samples and is usually assumed constant across people and removed by excluding the QRS complex. We quantified CFA in 7,995 polysomnography patients (98,203 channel-recordings) from the Human Sleep Project by regressing each channel's R-peak-locked EEG average on the ECG average outside a ±50 ms QRS interval. The ECG explained a mean R^2 of 0.32; removing the ECG-related independent component reduced heartbeat-locked variance by a median of 28%. CFA ranged from negligible to near-complete between patients and varied across electrodes. It increased with BMI ($r = 0.11$, $p = 4.3e-17$), was higher in men than women (0.37 vs. 0.32, $p = 1.5e-21$) independently of BMI, and rose with segment duration from 0.30 at 5 min to 0.43 at 90 min (Friedman $p < 1e-300$). Because its scalp pattern was consistent across patients, we developed CFAR (Cardiac Field Artifact Removal), an ECG-informed model that, once trained, removes CFA from EEG without ECG: in 6,711 held-out patients, it achieved 78% of the cleaning possible with the patient's own ECG (R^2 0.36 to 0.18, vs. 0.13) and cut the cardiac share of ongoing EEG variance from 1.8% to 0.6%, retaining 91% of a simulated neural HEP. EEG studies should estimate CFA per channel, report segment length, and account for BMI and sex; CFAR can clean recordings without ECG.

Highlights

- In 7,995 patients, the ECG explained 32% of heartbeat-locked EEG variance after QRS exclusion.
- CFA rose with BMI ($r = 0.11$, $p = 4.3e-17$) and segment duration (R^2 0.30 at 5 min, 0.43 at 90 min).
- CFA was higher in men (0.37 vs. 0.32), independent of BMI ($p = 3.2e-13$).
- Without ECG, CFAR achieved 78% of ECG-based cleaning, and removed more CFA in obese patients ($p = 0.001$).

Keywords: cardiac field artifact; CFAR; EEG artifact; ECG; body mass index; sex differences; segment duration; heartbeat-evoked potential; independent component analysis; polysomnography

1. Introduction

The heart is one of the body's strongest bioelectrical generators, and its field is volume-conducted to the scalp. Every EEG recording therefore contains a cardiac field artifact (CFA): a heartbeat-locked deflection of non-cortical origin[1-4]. CFA adds structured noise to any EEG measure[3,4] and is preserved, not averaged out, in R-peak-locked analyses. The heartbeat-evoked potential (HEP), used to study cardiac interoception[5-7], is the clearest example: CFA and cortical responses to the heartbeat share the same time-locking event[1,6]. The standard mitigation is to exclude ±30–50 ms around the QRS complex and, in some studies, to remove an ECG-related independent component[6-8].

CFA has been characterised only in single cohorts of a few tens of participants[1,2], too small to detect modest differences between people. It is therefore usually treated as a constant property of the recording, removed by a fixed QRS exclusion. Recent reviews note that heterogeneous CFA handling limits reproducibility and clinical interpretation[7-9]. If CFA differs systematically between people or analysis choices, a group difference in heartbeat-locked EEG may reflect cardiac electrophysiology, tissue conductivity, or electrode geometry rather than brain activity[2,6,10,11].

Three factors are of particular interest. First, body composition alters the conduction path from heart to scalp: subcutaneous fat attenuates surface-ECG voltage, correcting voltage for body fat changes its relation to cardiac structure[12], and obesity is associated with low QRS voltage and a shifted cardiac axis[13]. BMI, though an indirect measure of adiposity, is available in most clinical datasets. Second, men and women differ in thoracic anatomy, heart size, and fat distribution, so CFA may differ by sex, a common grouping variable. Third, CFA is estimated from averages over heartbeats, and averages over fewer events are less reliable[14], so apparent contamination may depend on segment duration. None of these factors has been examined at the population level.

We quantified CFA in clinical polysomnography recordings from 7,995 patients in the Human Sleep Project, a public, curated dataset[15], the largest sample in which CFA has been measured. We used two estimators: a model-free regression of each channel's R-peak-locked EEG average on the patient's R-peak-locked ECG average, and ECG-informed ICA. We show the range of CFA in individual patients, map its scalp distribution, and test its dependence on BMI, sex, age, and segment duration (5–90 min). We hypothesised that ECG-related variance would persist outside the QRS interval, increase with BMI, differ by sex, and increase with segment duration. Finally, we developed CFAR (Cardiac Field Artifact Removal), a model trained on the cohort's ECG-informed CFA patterns that removes CFA from EEG recorded without ECG.

2. Methods

2.1 Cohort and recordings

This is a secondary analysis of existing polysomnography recordings from the Human Sleep Project[15], a public, curated dataset of clinical recordings with linked electronic health records (EHR), distributed through the Brain Data Science Platform. BMI was computed from each patient's median EHR height and weight; values outside 10–80 kg/m² were excluded. EEG channels were identified by matching labels to 10-20/10-10 electrode names, which excluded intracranial and auxiliary channels; at least two were required. The ECG channel was the first label matching ECG/EKG. Cohort characteristics are in Section 3.1.

2.2 Window selection, R-peak detection, and preprocessing

For the main analysis, one 10-min window per recording was selected by a seeded random search (uniform start times; up to 10 attempts). R-peaks were detected on the ECG before the analysis filter: the median-subtracted signal was band-pass filtered at 5–25 Hz (third-order zero-phase Butterworth) and rectified, making detection polarity-independent, and peaks were identified with a minimum spacing of 300 ms and a prominence ≥3 × the median absolute deviation. A window was accepted if (i) ≥70% of EEG channels passed quality criteria (≥99.9% finite samples; SD 0.1–500 µV; peak-to-peak ≤5,000 µV; <20% flat samples), (ii) mean heart rate was 35–180 beats/min, and (iii) ≥70% of R-R intervals

were 0.33–2.0 s. EEG and ECG were then band-pass filtered from 1 Hz to min(100 Hz, Nyquist − 0.5 Hz) (zero-phase FIR, MNE-Python[16]; 86% of recordings were sampled at 200 Hz) and analysed in μV at native sampling rate, without re-referencing, interpolation, or amplitude-based rejection, so that CFA was estimated in minimally processed data. For the duration analysis, the procedure was repeated with 5-, 20-, 30-, 45-, 60-, 75-, and 90-min windows (Section 2.7). Parameters are summarised in Table 1.

Table 1. Window selection, quality control, and preprocessing parameters.

Step	Parameters
Window selection	Seeded random search, uniformly distributed start times, up to 10 attempts; 10 min (main analysis), 5–90 min (duration analysis, see Section 2.7)
R-peak detection	Median subtraction; 5–25 Hz third-order zero-phase Butterworth band-pass; rectification; minimum inter-peak distance 300 ms; prominence ≥3 × MAD
EEG channel quality	≥99.9% finite samples; SD 0.1–500 μV; peak-to-peak ≤5,000 μV; <20% flat samples
Window acceptance	≥70% of EEG channels pass quality criteria; mean heart rate 35–180 beats/min; ≥70% of R-R intervals 0.33–2.0 s
Analysis filter	Zero-phase FIR band-pass, 1 Hz to min(100 Hz, Nyquist – 0.5 Hz), MNE-Python; applied to EEG and ECG
Signal handling	Native sampling rate, μV; no re-referencing, channel interpolation, or amplitude-based rejection
HEP epochs	–300 to 400 ms around each R-peak; no baseline correction; ≥20 epochs; ±50 ms QRS exclusion for outside-QRS R^2 (see Section 2.3)

2.3 Model-free CFA estimator: HEP-vs-ECG regression

HEP epochs spanned −300 to 400 ms around each R-peak, without baseline correction; epochs beyond the window edges were discarded, and recordings with fewer than 20 epochs were excluded (median 940 epochs per 10-min window). For each patient and channel, the R-peak-locked EEG average was compared with the ECG average over the same epochs. The squared zero-lag Pearson correlation (R^2) between the two waveforms, the fraction of HEP variance shared with the ECG, was the CFA index. R^2 was computed over the full epoch and outside a ±50 ms QRS-exclusion window[8]. This estimator does not depend on ICA isolating a cardiac source; it tests directly how far the averaged scalp waveform is a scaled, offset copy of the averaged ECG. For patient-level analyses (Section 2.7), R^2 was averaged across the patient's channels at the six well-covered sites (Section 2.5) and across recordings (patient-mean CFA R^2).

2.4 Model-based CFA estimator: ECG-informed ICA

In parallel, ICA[17] (extended Picard[18]; min(15, channels − 1) components; 500 iterations; fixed seed) was fitted to the filtered EEG of the same window. ECG-related components were identified by correlation with the ECG channel (MNE-Python default threshold[16]); if none exceeded it, the highest-scoring component was used (two components were flagged in 49 of 13,624 recordings). Each component's mixing-weighted contribution to each channel was evaluated on the R-peak-locked average, quantifying the fraction of heartbeat-evoked signal it explains. Removal was also measured directly, as the reduction in each channel's HEP variance after back-projection without the component, 100 × (pre − post)/pre. For Figure 2b, the ratio of component to residual variance (SNR) was computed per channel-recording.

2.5 Channel canonicalisation and minimum coverage

Channel labels were mapped to their scalp-side 10-20 site; reference-only labels were excluded. Figure 2c–d includes sites recorded in at least 500 patients. Analyses comparing conditions or estimators at common electrodes (Sections 3.4–3.6, S2–S4) used sites present in ≥50% of patients, which retained six sites (F3, F4, C3, C4, O1, O2; each in 96–100% of patients), the dataset's predominant montage.

2.6 EEG-ECG cross-correlation and mutual information

The EEG-ECG relationship was further characterised in a separate subsample of 150 patients (Figures S3–S4; 148 with usable data). Lag-resolved cross-correlation was the Pearson correlation between each channel's evoked waveform and the ECG evoked average at lags of ±100 ms in 5-ms steps; the peak absolute correlation and its lag were reported after per-channel sign alignment, because reference polarity is arbitrary. Mutual information was estimated with the Kraskov k-nearest-neighbour estimator[19] to capture linear and nonlinear dependence. Both were computed pre-ICA, post-ICA, and for a non-heartbeat-locked control in which the same window was re-epoched around an equal number of random pseudo-events; in this subsample, ICA was refitted and R-peaks were re-detected on the filtered ECG. For the power spectral density (PSD) comparison (Figure S4), the ECG evoked average was added as a fourth condition, and Welch PSD (dB) was computed at the six well-covered electrodes. Two further random subsamples supported Figure S2: in one (973 patients), ICA windows were re-epoched around pseudo-events, without refitting, to estimate a variance noise floor; in the other (492 patients), ICA was refitted and the spectral entropy of each evoked waveform (normalised Shannon entropy of its Welch spectrum) was computed for the three conditions.

2.7 Statistics

Diagnoses were assigned to 15 predefined categories by case-insensitive keyword matching on EHR diagnosis descriptions (Table S1); categories overlap. EHR-linked patients without a recorded diagnosis formed the no-diagnosis group, which included patients from a site without diagnosis tables (see Limitations); patients without an EHR link were excluded from diagnosis analyses. Age was tested as a continuous variable (Pearson correlation) and by tertile (one-way ANOVA). Sex and presence of a diagnosis were tested with the Mann-Whitney U test. Diagnosis categories (Section 3.6) were compared with the Kruskal-Wallis test and pairwise Mann-Whitney tests, corrected by the Benjamini-Hochberg false discovery rate (FDR)[20]. BMI was tested with Pearson correlation, using the longest available window per patient (30 min for 98%) to keep one observation per patient; absolute R^2 in the BMI and BMI-adjusted sex analyses is therefore not comparable with the 10-min estimates. BMI correlations were also computed by sex, and a BMI-by-sex interaction term in an ordinary least-squares (OLS) model tested whether slopes differed. A further OLS model compared the sex coefficient with and without BMI adjustment. Model-free R^2 and log ICA SNR (Section 2.4) were correlated across channel-recordings (Pearson; Figure 2b). Segment duration (Section 3.5) was tested in patients with data at all eight durations (5–90 min), using all EEG channels (Figure 3d shows channel-level means). Patient-mean R^2 was compared across durations with the Friedman test (effect size, Kendall's W), and consecutive durations, 5 vs. 90 min, and 10 vs. 90 min with Holm-corrected paired Wilcoxon signed-rank tests. Sexes and diagnosis groups were compared at each duration with Mann-Whitney tests. Windows of each length were drawn with the same seeded procedure, so shorter windows were usually nested within longer ones; the 45- to 90-min analyses used one recording per patient. Sample sizes differ because analyses need different data: model-free estimator, all patients with a valid window (7,995); ICA, patients with a converged decomposition (7,933); age, sex, and diagnosis, EHR-linked patients with known age and at least one well-covered site (7,729); BMI, patients with a plausible BMI (6,127); duration, patients with data at all eight durations (6,768). For illustration (Figure 1), four patients with measured BMI and at least 12 electrodes in a 60-min segment were chosen: a pair matched on sex, age (±2 years), and BMI (±1 kg/m²) with the largest CFA difference, and a pair differing in sex, age (≥35 years), and BMI (≥15 kg/m²) with near-equal CFA; they were not used for inference. To test sleep stage (Section 3.1), a random subsample of 23 patients was staged in 30-s epochs with YASA[21]; for wake, light sleep (N1+N2), N3, and REM, up to 2 windows per duration (2, 5, 10, 20, 30, 60 min) were drawn from contiguous stretches of each stage, and CFA R^2 was computed as in Section 2.3. Stage and duration were compared with one-way ANOVA.

2.8 CFAR: ECG-free cardiac field artifact removal

Many EEG datasets lack an ECG. We therefore developed CFAR (Cardiac Field Artifact Removal), an ECG-informed model: the ECG is needed to train it, not to apply it. Single-beat CFA was too weak to detect heartbeats from the EEG, but CFA has a largely fixed scalp pattern, which allows cleaning without locating beats. In recordings with the standard six-channel, mastoid-referenced montage (F3, F4, C3, C4, O1, O2), each patient's CFA pattern (first singular vector of the R-peak-locked EEG average) was computed, sign-aligned, and averaged, weighted by the patient's CFA R^2, into one population pattern. All 6,711 eligible patients were used with 5-fold cross-fitting: each patient was cleaned with a pattern

learned from the other folds (about 5,369 patients), projected out of every sample of the continuous EEG (signal-space projection). The ECG was used only for scoring: CFA R^2 (outside QRS) before and after cleaning, compared with patient-specific CFAR (below). Cardiac noise in the ongoing EEG was quantified as the share of continuous EEG variance carried by the ECG-locked template overlap-added at every beat. To measure preservation of brain activity, a synthetic neural HEP (Gaussian, −2 μV at +300 ms, frontal > central > occipital) was added before cleaning and the retained fraction was measured; the cost to ongoing EEG was the change in Welch power (1–40 Hz). To test whether participant information helps, a ridge regression trained in the same folds predicted each patient's pattern from BMI (median-imputed, with a missing indicator), age, sex, number of diagnoses, and diagnosis categories.

Patient-specific CFAR. When the ECG is available, the same projection can use the patient's own CFA pattern instead of the population one: the ECG R-peaks give the R-peak-locked EEG average, its first singular vector is the patient's CFA pattern, and that pattern is projected out of every sample. This is the best a single spatial direction can do for a given patient and so bounds what population CFAR can reach. The pattern is learned from the same window it is scored on, so its performance is an optimistic estimate.

Comparison of cleaning pipelines (Figure 4d). In 6,711 patients, CFAR and patient-specific CFAR were compared with ECG-informed ICA (Section 2.4), with regression on the ECG (each channel regressed on the ECG at lags of ±20 ms plus an intercept, and the fit subtracted), and with CFAR followed by ECG-informed ICA fitted on the CFAR output (CFAR+ICA). The CFAR filter was cross-fitted (5 folds, unweighted mean of patterns), so each patient was cleaned by a filter learned from other patients. Two scores were computed per patient. ECG-explained EEG variance left was CFA R^2 multiplied by the variance of the R-peak-locked average outside the QRS, averaged over channels and expressed as a percentage of the raw EEG value; unlike R^2, which measures only how ECG-like the residual is, it falls when the artifact shrinks. Neural signal preserved was the share of the simulated heartbeat-locked brain response (above) recovered after cleaning. Medians are reported with 95% bootstrap confidence intervals.

3. Results

3.1 Cohort

Demographics were available for 7,732 of the 7,995 patients (median age 61 years, range 7–103; Male n=3,965, Female n=3,767). The EHR-linked cohort was clinically referred, with a high diagnostic burden; the most prevalent categories were Obstructive Sleep Apnea (n=5,524); Hypertension (n=3,882); Obesity (n=3,016) (Figure S1).

Windows were placed at random across the recording (Section 2.2), so they sampled wake and sleep stages roughly in proportion to their duration, and cohort estimates describe CFA over the whole night. In 23 patients staged with YASA[21] (Section 2.7), mean CFA R^2 was similar across stages (wake 0.30, light sleep 0.31, N3 0.32, REM 0.32; ANOVA p = 0.76), whereas segment duration had a significant effect in the same data (p = 2.6e-09). The main conclusions are therefore not specific to one sleep stage.

3.2 CFA in individual patients

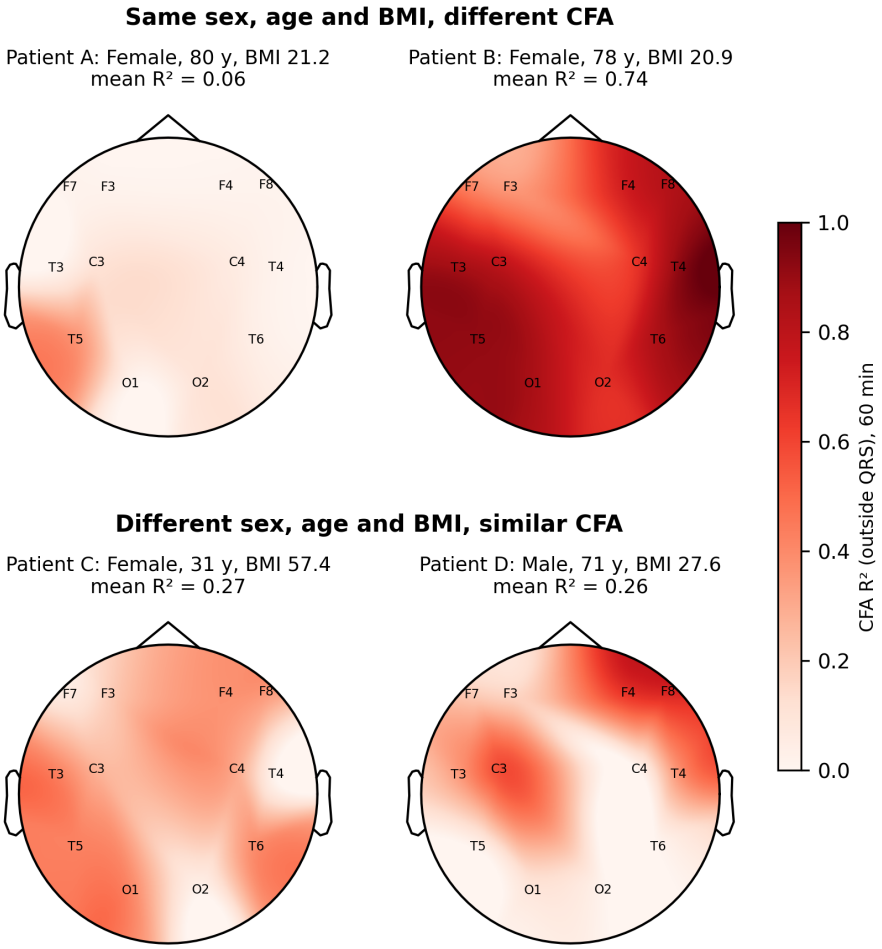


Figure 1. CFA in four example patients. Per-electrode CFA R^2 (outside QRS) from a 60-min segment. Top: patients matched on sex, age, and BMI with very different CFA. Bottom: patients differing in sex, age, and BMI with similar CFA. Patient A: female, 80 years, BMI 21.2 kg/m², mean R^2 0.06. Patient B: female, 78 years, BMI 20.9 kg/m², mean R^2 0.74. Patient C: female, 31 years, BMI 57.4 kg/m², mean R^2 0.27. Patient D: male, 71 years, BMI 27.6 kg/m², mean R^2 0.26.

Figure 1 shows that CFA is a property of the individual that demographics alone do not predict. Patients A and B were both female, aged 80 and 78 years, with BMI 21.2 and 20.9 kg/m², yet in a 60-min segment the ECG explained 6% of heartbeat-locked EEG variance in Patient A and 74% in Patient B, spread over most of the scalp. Conversely, Patient C (female, 31 years, BMI 57.4 kg/m²) and Patient D (male, 71 years, BMI 27.6 kg/m²) differed in all three, but CFA was similar (27% and 26%) with a different spatial pattern. The analyses below test how CFA relates to BMI, sex, age, and segment duration across the cohort.

3.3 CFA across the cohort

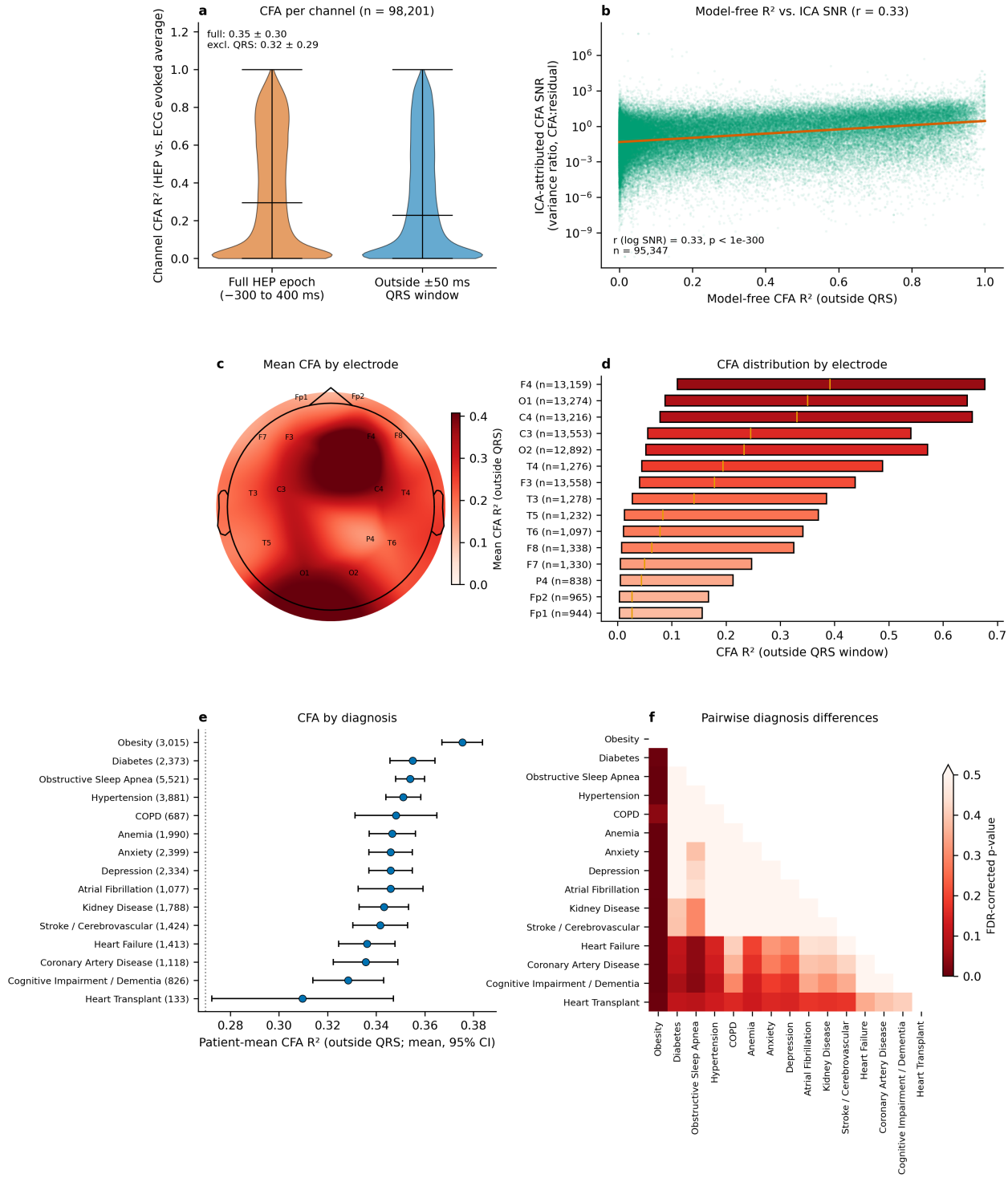


Figure 2. CFA across 7,995 patients (10-min segments). (a) Per-channel R² between R-peak-locked EEG and ECG averages, full epoch and outside QRS. (b) ICA SNR vs. R² (log axis; r = 0.33, p < 1e-300). (c) Mean CFA R² per 10-20 site. (d) Per-site R² distribution (box, interquartile range). (e) Patient-mean CFA R² per diagnosis category (mean ± 95% CI); dotted line, no recorded diagnosis; Kruskal-Wallis p = 9.1e-21. (f) Pairwise Mann-Whitney q-values (Benjamini-Hochberg).

Outside the QRS-exclusion window, the ECG explained a mean R² of 0.32 (median 0.23; Figure 2a), only modestly below the full-epoch mean (0.35), so QRS exclusion removed only about 10% of the ECG-explained variance. Values ranged from no detectable CFA to EEG averages almost identical to the ECG. In the 148-patient subsample with a

chance-level reference (Section 2.6), full-epoch R^2 at F3, F4, C3, and C4 was 0.35 for R-peak-locked averages but 0.04 (median 0.03) around random pseudo-events, so the observed R^2 far exceeded chance.

The model-based estimator agreed: the ECG-related ICA component accounted for a median of 21% of heartbeat-locked EEG variance, and excluding it reduced this variance by a median of 28%. Channels with higher model-free R^2 had higher ICA SNR (Figure 2b); the dispersion is expected, because one globally selected source does not load equally on all electrodes. Regression on the concurrent ECG has been proposed for estimating and removing CFA, in linear[22] and nonlinear neural-network[23] form; we therefore used per-channel ECG regression as the primary estimate and ICA as convergent evidence.

CFA varied across the scalp (Figure 2c–d), from 0.41 at F4 to 0.12 at Fp1, a 3.3-fold range. Sites outside the standard montage came from a minority of recordings with other montages and references, so part of this range reflects referencing rather than position; within the six standard mastoid-referenced sites, mean R^2 ranged from 0.27 (F3) to 0.41 (F4). A site-independent correction would therefore perform unevenly.

3.4 CFA increases with BMI and is higher in men

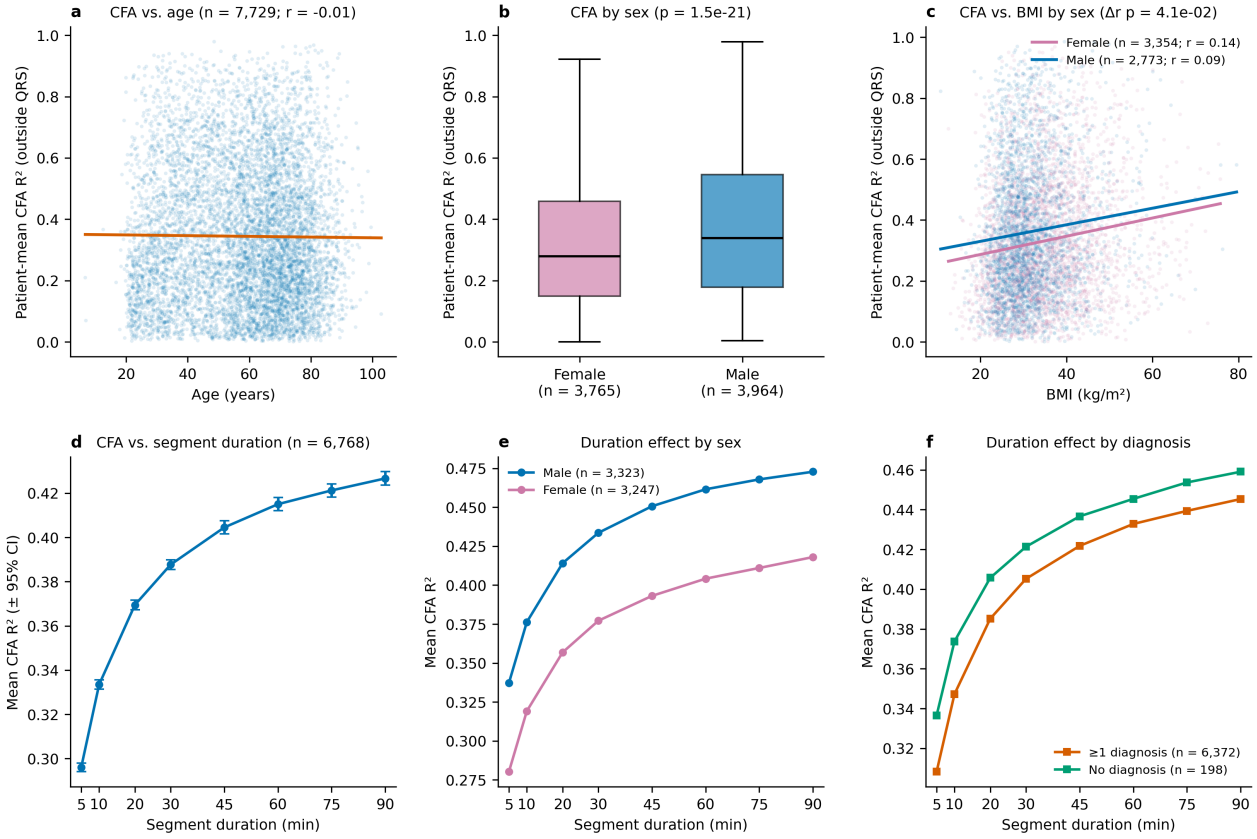


Figure 3. CFA vs. patient characteristics (a–c) and segment duration (d–f). Patient-mean CFA R^2 (outside QRS) vs. (a) age (r = -0.01, p = 0.41), (b) sex (Mann-Whitney p = 1.5e-21), and (c) BMI by sex (female r = 0.14, male r = 0.09; difference, Fisher z p = 0.041). (d–f) Patients with data at all eight durations (n = 6,768). (d) Mean R^2 ± 95% CI; Friedman p < 1e-300. (e) By sex. (f) By diagnosis.

BMI correlated positively with CFA R^2 (Figure 3c; n = 6,127, r = 0.11, p = 4.3e-17; mean BMI 33.5 kg/m²): each 10 kg/m² added 0.025 to patient-mean R^2. The correlation held in women (r = 0.14, p = 4.0e-17) and men (r = 0.09, p = 9.6e-07), with no BMI-by-sex interaction (p = 0.67): CFA rose with BMI at the same rate in both sexes. The correlation was nonetheless stronger in women (Fisher z p = 0.041), meaning BMI explained more of the between-patient variation in women, not that CFA rose faster with BMI. Though weak individually ($r^2 \approx 0.01$), it agrees with the diagnosis analysis, in which obese patients had the highest CFA of all categories (Section 3.6).

CFA R^2 was higher in men than in women (0.37 vs. 0.32, p = 1.5e-21; Figure 3b). BMI did not explain this: women had a higher mean BMI (34.8 vs. 31.8 kg/m²), and the sex coefficient was unchanged after BMI adjustment (0.031 vs. 0.040, p = 3.2e-13; Section S5), with BMI an independent predictor (p = 1.5e-21). The sex difference held at every duration (Section 3.5). Age showed no linear relationship with CFA (Figure 3a; r = -0.01, p = 0.41); only the oldest tertile was slightly lower (0.33 vs. 0.35/0.35, p = 1.0e-04).

3.5 Segment duration changes the estimated CFA

Segment duration had a significant, systematic effect (Friedman χ^2 = 10,184, p < 1e-300, Kendall's W = 0.21; n = 6,768). Mean CFA R^2 rose from 0.30 at 5 min to 0.33 (10 min), 0.37 (20), 0.39 (30), 0.40 (45), 0.42 (60), 0.42 (75), and 0.43 at 90 min (Figure 3d), with every consecutive step significant after Holm correction. From 5 to 90 min, R^2 increased in 80% of patients (mean +0.14); from the 10-min main-analysis windows to 90 min, in 76% (mean +0.10). Gains shrank with duration, from 0.036 between 10 and 20 min to 0.005 between 75 and 90 min, converging to R^2_∞ = 0.451 (saturating fit). Fewer channel-recordings contributed at 45–90 min (44,044 vs. 76,045 at 30 min), which may change channel composition, but between 5 and 30 min composition was nearly constant (76,343 to 76,045 channel-recordings) and R^2 still rose monotonically.

Men had higher CFA than women at every duration (Figure 3e; e.g. 0.34 vs. 0.28 at 5 min, 0.47 vs. 0.42 at 90 min). Diagnosis differences were not significant (Figure 3f); the no-diagnosis mean was numerically higher at every duration, consistent with the within-site reversal in the Limitations.

3.6 CFA by diagnosis category

Among diagnosis categories (Figure 2e–f), CFA was highest in Obesity (+0.11 R^2 vs. the no-diagnosis reference, n = 3,015) and lowest in Heart Transplant (+0.04, n = 133). Of 105 pairwise comparisons, 17 survived FDR correction; 14 involved Obesity, which differed from every other category, and the rest involved Obstructive Sleep Apnea, in line with the BMI association (Section 3.4). All 15 categories lay above the no-diagnosis reference, but this reference is confounded by site (see Limitations) and comparisons were not adjusted for age, sex, or BMI, so these are not disease effects.

3.7 CFAR removes CFA without ECG

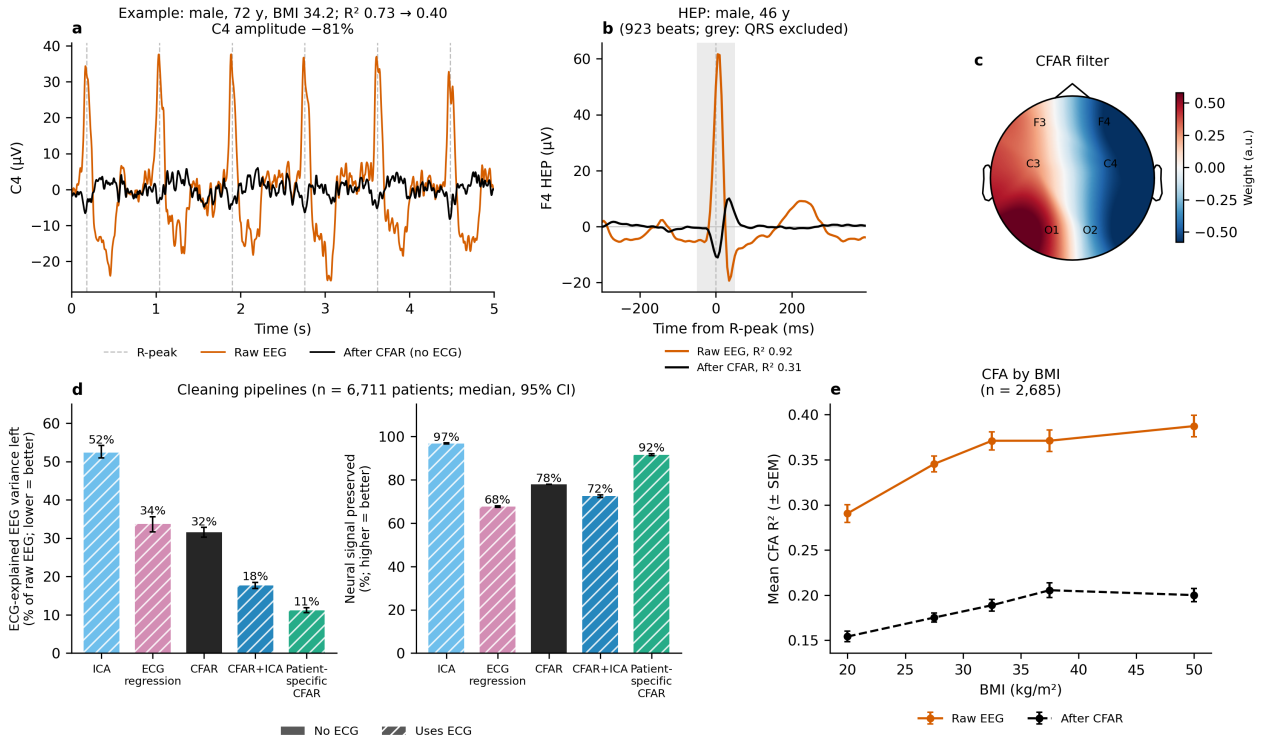


Figure 4. CFAR: ECG-free cardiac field artifact removal (Section 2.8). (a) Five seconds of EEG at C4 from an example patient: raw (orange) and after CFAR (black); dashed lines, R-peaks. (b) Heartbeat-evoked potential (R-peak-locked average, C4) of a second high-CFA patient before and after CFAR; grey, QRS window excluded from R^2. (c) CFAR filter: the CFA scalp pattern learned from the cohort. (d) Cleaning pipelines in 6,711 patients (CFAR filter cross-fitted, so each patient is cleaned by a filter learned from other patients), sorted by artifact left. Left panel, ECG-explained EEG variance left (CFA R^2 × heartbeat-evoked variance outside the QRS, % of raw EEG). Right panel, neural signal preserved: % of a simulated heartbeat-locked brain response (2 µV, 300 ms after each R-peak) recovered after cleaning. Medians with 95% bootstrap CIs; hatched, ECG used (ICA component selection, ECG regression, or the patient's own ECG-derived pattern for patient-specific CFAR). (e) Mean CFA R^2 (± SEM) by BMI across all patients with known BMI, before (solid) and after (dashed) CFAR.

The CFA scalp pattern was highly consistent across patients (median absolute cosine similarity to the population pattern, 0.93; Figure 4c). CFAR, which projects this pattern out of held-out recordings without ECG, reduced mean CFA R^2 from 0.36 to 0.18 (49% reduction; Wilcoxon $p < 1e\text{-}300$), and R^2 fell in 79% of patients. Patient-specific CFAR reached 0.13, so CFAR closed 78% of the gap. In the example patient, mean R^2 fell from 0.73 to 0.40, and the deflection after every R-peak was removed (Figures 4a-b). Because R^2 measures the ECG-likeness of the residual rather than its size, we compared pipelines in 6,711 patients on the ECG-explained EEG variance left (CFA R^2 × heartbeat-evoked variance outside the QRS, relative to raw EEG) and on neural signal preserved (the share of a simulated heartbeat-locked brain response recovered after cleaning; Figure 4d). ECG-informed ICA left a median 52% of the ECG-explained variance, CFAR 32% and regression on the ECG 34%, preserving 97%, 78% and 68% of the neural signal. Applying ICA after CFAR left 18% (preserving 72%), and patient-specific CFAR, which needs the patient's own ECG, left 11% (preserving 92%). Obese patients (BMI ≥ 30 kg/m², n = 1,525) had more CFA removed than lean patients (BMI < 25, n = 442; R^2 –0.18 vs. –0.14, Mann-Whitney $p = 0.0011$), because they had more to remove: the share removed was similar (48% vs. 47%). Residual CFA remained higher at high BMI (Figure 4e). In the ongoing EEG, the cardiac share of variance fell from a median of 0.70% to 0.25% (mean 1.8% to 0.6%; median reduction 53%) and decreased in 89% of patients. Predicting each patient's pattern from BMI, age, sex, and diagnoses gave no gain (median cosine similarity to the patient's own pattern 0.933 vs. 0.932; mean R^2 0.180 vs. 0.180; $p = 0.011$; better in only 48% of patients): these factors change the amount of CFA, not its scalp pattern, and projection removes the pattern regardless of amount. Cleaning retained a median 91% of a synthetic neural HEP and changed ongoing EEG power by a median of -1.6 dB (1–40 Hz), the cost of removing one of six spatial dimensions.

4. Discussion

In 7,995 patients, by far the largest CFA sample, two estimators showed that much of the heartbeat-locked scalp EEG variance is shared with the ECG even after QRS exclusion. CFA was not a generic nuisance but a property of each person and analysis: it ranged from negligible to near-complete between patients (Figure 1), varied 3.3-fold across sites (Figure 2c–d), increased with BMI, was higher in men (Figure 3a–c), and depended strongly on segment duration (Figure 3d–f). Because these are common grouping variables or analysis choices, uncorrected CFA can create, mask, or inflate group differences in heartbeat-locked EEG.

The conventional ±30–50 ms QRS exclusion[6-8] did not remove CFA: excluding ±50 ms reduced mean R^2 only from 0.35 to 0.32, so ECG-correlated signal extends into the analysis interval. The agreement between model-free R^2 and ICA SNR shows that both detect the same contamination. Neither proves that every residual heartbeat-locked deflection is artifactual, and cortical heartbeat-evoked activity is well documented[5,6,10]; still, uncorrected heartbeat-locked EEG variance cannot be assumed cortical.

BMI correlated positively with CFA in women and men, and obese patients had the highest CFA of all diagnosis categories. This seems counterintuitive: adipose tissue conducts roughly ten times less than muscle[24], and subcutaneous fat attenuates the surface ECG, so obesity is associated with low QRS voltage[12,13]. If fat only insulated, less cardiac field should reach the scalp. The resolution lies in what R^2 measures. R^2 is scale-invariant: it indexes how closely the shape of the heartbeat-locked scalp average follows the ECG, not amplitude. Uniform attenuation scales both without changing their correlation; R^2 rises only when the cardiac component grows relative to everything else in the average, namely background EEG and neural responses. Several changes in obesity could raise this share. First, obesity is associated with greater left-ventricular mass and a leftward, more horizontal cardiac axis[13], changing the strength and orientation of the cardiac dipole; in mastoid-referenced derivations, CFA is the cardiac potential difference between scalp and mastoid, which depends strongly on dipole orientation. Second, low-conductivity tissue redistributes volume currents as well as attenuating them, which may change how much of the field projects onto the head. Third, tissue between brain and electrode attenuates cortical EEG at the same electrode. More heartbeats per window is unlikely to explain the effect: heart rate was slightly lower at higher BMI ($r = -0.07$), and adjusting for heart rate did not reduce the BMI coefficient (0.0024 vs. 0.0025 per kg/m², $n = 5{,}985$). Which mechanism dominates cannot be resolved here; this needs absolute CFA amplitude, which should fall with BMI if insulation dominated, and direct measures of body composition and heart position. Individually the correlation was weak ($r = 0.11$), so BMI cannot predict a recording's CFA, but a difference in mean BMI between study groups, common in clinical comparisons, will produce a systematic difference in CFA.

CFA was higher in men at every segment duration, unchanged by BMI adjustment; because women here had a higher mean BMI, BMI would if anything have reduced the difference. It may reflect differences in heart size and position, thoracic and head geometry, or fat distribution, none measured here; cardiac parameters such as stroke volume may also influence heartbeat-locked responses[11]. Sex differences are among the most reported group effects in EEG, so studies comparing men and women on heartbeat-locked measures should quantify CFA per channel and show that the difference survives its correction.

Segment duration had the largest, most consistent effect: in 6,768 duration-matched patients, mean R^2 rose from 0.30 at 5 min and 0.33 at 10 min to 0.43 at 90 min, every step was significant, and R^2 increased from 5 to 90 min in 80% of patients. Longer segments contain more heartbeats and give a more stable average[14], letting shared EEG-ECG structure emerge from background EEG. Two consequences follow. First, short segments, including our 10-min main analysis, underestimate CFA, so a low estimate from a short recording does not show that the data are clean. Second, groups that differ in recording length or usable heartbeats will differ in apparent CFA for methodological reasons alone. Segment length should therefore be reported, justified, and matched between groups where possible, and results checked across durations. Because R^2 depends on the number of averaged beats, group differences in heart rate could also affect R^2; this was not examined.

BMI is an anthropometric index, not a direct measure of adiposity; fat mass, thoracic geometry, and electrode impedance were unavailable, and diagnosis categories overlap. We therefore treat BMI, obesity, and sex as confounders of CFA, not

as evidence of a causal effect of body fat. Associations of age and diagnosis with CFA may likewise reflect anatomy, cardiac physiology, medication, comorbidity, or recording conditions.

In practice, EEG preprocessing should combine concurrent ECG, channel-level assessment of the R-peak-locked EEG-ECG relationship, and removal or regression of ECG-related components, then verify that heartbeat-locked variance falls after cleaning while staying above a non-heartbeat-locked noise floor. Reporting only the QRS mask or the selected ICA component is insufficient. Studies should report segment duration and number of heartbeats, match them between groups where possible, report channel-level CFA before and after cleaning, and test robustness to adjustment for BMI (or, better, direct body-composition measures) and sex. These recommendations extend recent HEP reviews[8,9] to EEG analysis generally.

For recordings without ECG, CFAR offers a practical route. Because the CFA scalp pattern is consistent across people, CFAR, learned once from patients with ECG, cleaned the heart signal from the continuous EEG of unseen patients without a cardiac reference: it removed about 49% of heartbeat-locked CFA, cut the cardiac share of ongoing EEG variance by about 68%, and closed 78% of the gap to cleaning with the patient's own ECG. Because it acts on every sample, it cleans ongoing EEG, not only heartbeat-locked averages, and can be applied retrospectively to existing datasets. Conditioning on BMI, sex, age, and diagnoses gave no benefit, because these change the amount of CFA, not its pattern; they remain useful for identifying who needs cleaning most, since CFA before and after cleaning was highest at high BMI (Figure 4e) and in men (Figure 3). CFAR is specific to the montage and reference it was trained on, removes one of six spatial dimensions, retained 91% of a simulated neural HEP, and leaves some CFA; it should be validated on other montages before routine use.

5. Conclusions

In 7,995 patients, CFA was a substantial, systematic component of scalp EEG[1-4]: the ECG explained a mean R^2 of 0.32 of heartbeat-locked EEG variance outside the QRS window, and ICA removal of the ECG-related component reduced this variance by a median of 28%. CFA increased with BMI, was higher in men independently of BMI, varied across electrodes, and increased with segment duration up to 90 min. It is therefore a participant-, channel-, and duration-dependent noise source. Robust inference requires ECG-informed channel-level cleaning with before-and-after validation[8,9], reported and justified segment lengths[14], and adjustment for BMI and sex; without ECG, CFAR removes about half of the CFA. Otherwise, group differences in EEG may partly reflect the heart rather than the brain[6,11].

Limitations

First, diagnosis records covered only part of the Human Sleep Project, so the no-diagnosis group came largely from a site without diagnosis tables (mean R^2 0.19, n = 319). Within the site with records, patients without a diagnosis had higher R^2 than those with one (0.41, n = 181, vs. 0.35, n = 7,229). The with/without-diagnosis comparison is therefore confounded by site and is not a disease effect; comparisons among categories are unaffected. Second, detected heart rate was high for sleep (median 94 beats/min, interquartile range 78–111); the custom R-peak detector was not validated against an established algorithm, and occasional T-wave detections would mix T-wave-locked epochs into the averages. Third, R^2 cannot separate volume-conducted CFA from neural activity with an ECG-like time course, and epochs ended 400 ms after the R-peak, so later HEP intervals overlapping the T wave were not covered. Fourth, CFA was estimated in native references, mostly contralateral mastoid (e.g., F3-M2, F4-M1); because the cardiac field projects differently onto each reference, site values and hemispheric asymmetries may not generalise to average-referenced data.

Fifth, 10-min estimates are likely lower bounds, because CFA R^2 rose monotonically with duration, including from 5 to 30 min, where channel composition was nearly constant (Section 3.5). Sixth, chance-level (pseudo-event) R^2 was estimated only in 148 patients and only over the full epoch (Section 3.3). Seventh, windows were not selected by sleep stage, so vigilance was not controlled and may differ between groups and durations; CFA did not differ by stage in a small staged subsample (n = 23; Section 3.1). Eighth, the cohort was clinically referred, with a high diagnostic burden (Figure S1c), which limits generalisation of absolute R^2 to healthy volunteers, and underweight patients were

under-represented (55 of 6,127 with BMI <18.5 kg/m²). Finally, as a secondary analysis, recording parameters and clinical annotations could not be controlled.

Ethics statement

Human Sleep Project data were de-identified under the HIPAA Safe Harbor standard and made available under an approved institutional review board protocol (IRB protocol #2022P000417) with a waiver of informed consent; they were accessed under the Brain Data Science Platform data use agreement. No attempt was made to re-identify participants.

Data availability statement

The Human Sleep Project dataset is publicly available, subject to a data use agreement, from the Brain Data Science Platform (`https://bdsp.io/content/hsp/`). Analysis code is available from the corresponding author upon reasonable request (nircafri@mail.tau.ac.il).

Funding

Declaration of competing interests

The authors declare no competing interests.

Supplementary analysis

S1. Cohort composition

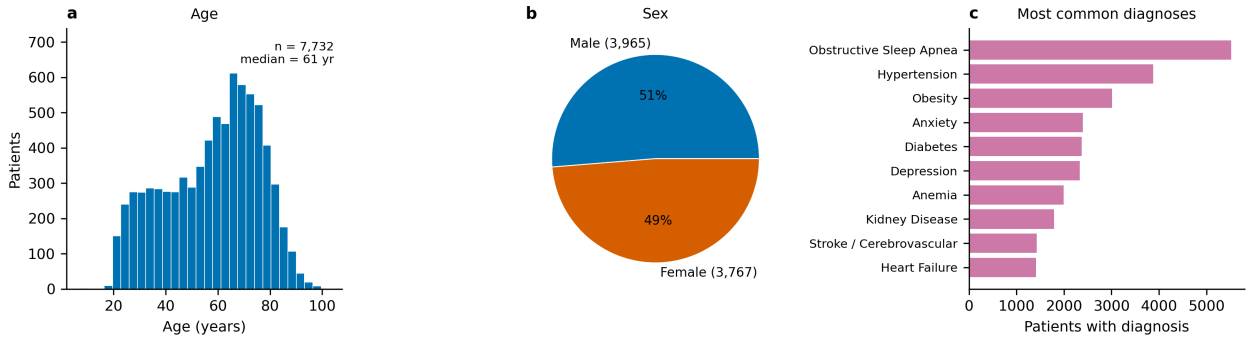


Figure S1. Cohort composition (7,738 Human Sleep Project patients). (a) Age. (b) Sex. (c) The ten most prevalent diagnosis categories (overlapping).

Table S1. Diagnosis categories and the case-insensitive keywords matched against EHR diagnosis descriptions. Keyword matching is inclusive; for example, "infarction" also matches myocardial infarction.

Category	Keywords (any match)
Obstructive Sleep Apnea	sleep apnea
Hypertension	hypertension
Diabetes	diabetes
Heart Failure	heart failure
Coronary Artery Disease	coronary; atherosclerotic heart
Heart Transplant	heart transplant; transplant status
Atrial Fibrillation	atrial fibrillation; atrial flutter
COPD	chronic obstructive
Obesity	obesity; obese
Depression	depressive; depression
Anxiety	anxiety
Cognitive Impairment / Dementia	cognitive; alzheimer; dementia
Stroke / Cerebrovascular	infarction; cerebrovascular; stroke; hemorrhage
Kidney Disease	kidney; renal
Anemia	anemia

S2. Variance and entropy vs. a non-heartbeat-locked noise floor

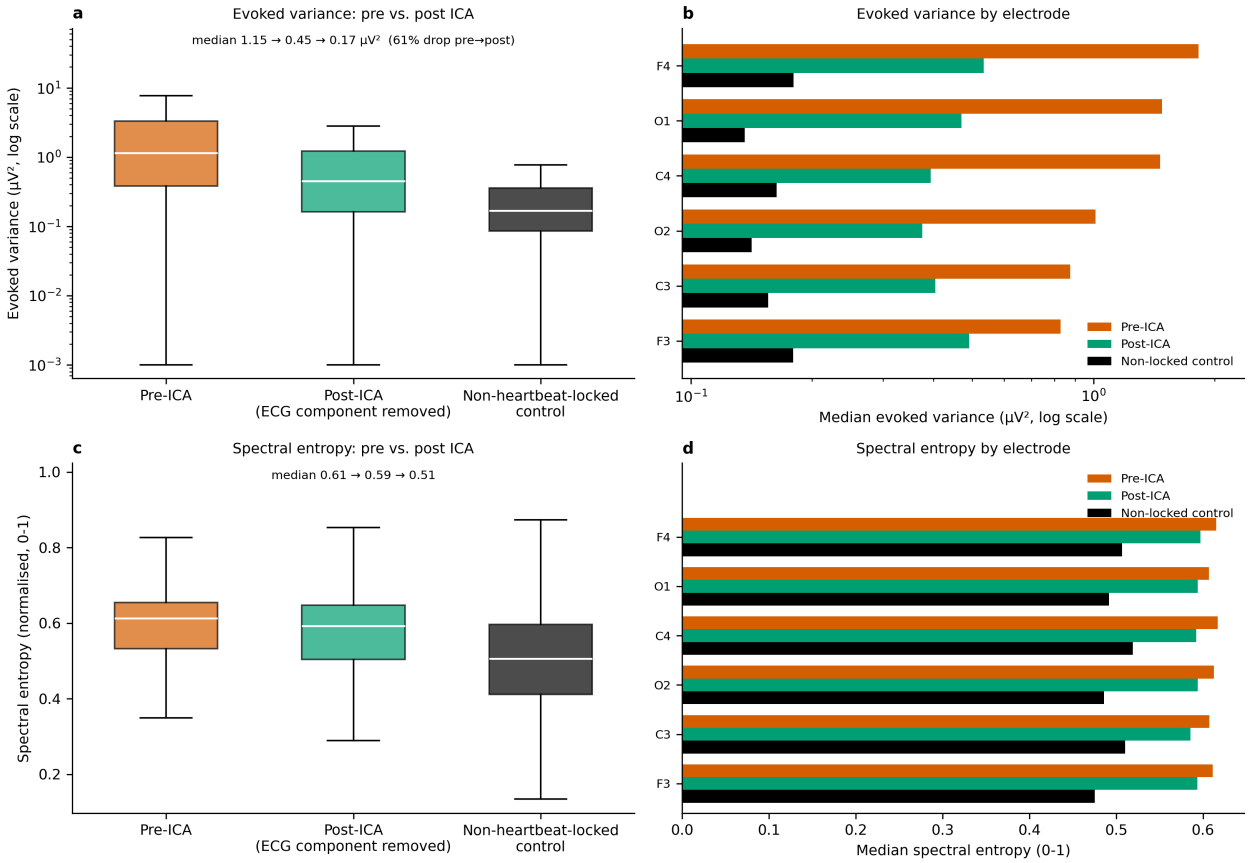


Figure S2. HEP variance before and after excluding the ECG-related ICA component, vs. a non-heartbeat-locked control (medians, log axis). (a) Core electrodes (F3/F4/C3/C4). (b) Per electrode. (c) Spectral entropy, core electrodes. (d) Spectral entropy per electrode.

Removing the ECG-related component reduced median core-electrode HEP variance by 61%; this ratio of group medians is larger than, and not comparable to, the median per-channel reduction in Section 3.3 (28%). The control's noise floor (0.17 µV²) was well below both pre-ICA (1.15 µV²) and post-ICA (0.45 µV²) values, and post-ICA variance stayed above it at every electrode (panel b), so component removal did not reduce the heartbeat-locked signal to noise. Spectral entropy, a variance-independent measure, fell from pre-ICA (0.61) to post-ICA (0.59) to the control (0.51); the post-ICA decrease was small, and post-ICA values exceeded the control at every electrode (panel d).

S3. EEG-ECG cross-correlation and mutual information

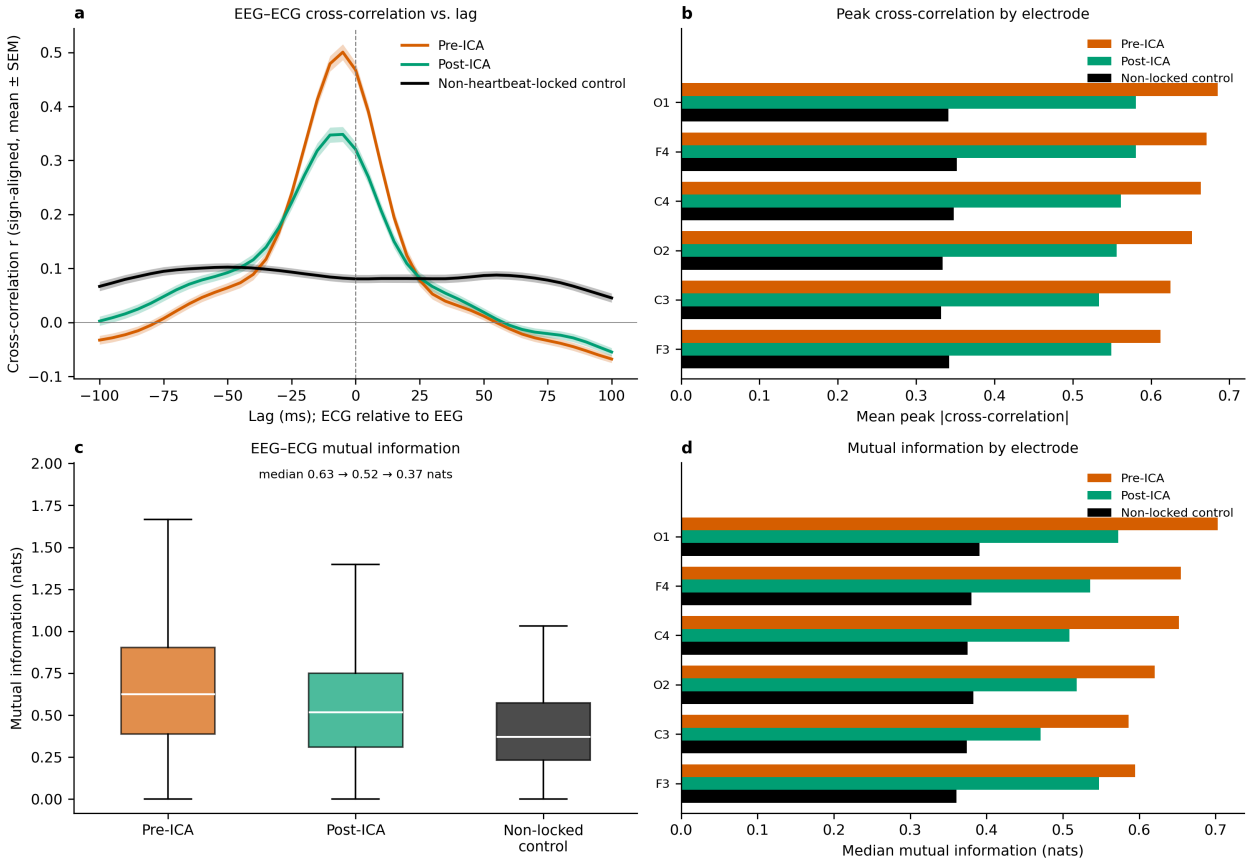


Figure S3. EEG-ECG cross-correlation and mutual information pre-ICA, post-ICA, and for a non-heartbeat-locked control (n = 148). (a) Cross-correlation vs. lag, core electrodes (F3/F4/C3/C4); mean ± SEM. (b) Mean peak |cross-correlation| per electrode. (c) Mutual information, core electrodes. (d) Mutual information per electrode.

Cross-correlation peaked near zero lag in both heartbeat-locked conditions (median -5 ms pre-ICA, -10 ms post-ICA), consistent with the instantaneous volume conduction assumed by the zero-lag estimator (Section 2.3). Peak |r| fell from 0.64 pre-ICA to 0.56 post-ICA and 0.34 for the control, and mutual information from 0.63 to 0.52 to 0.37 nats (panel c). Post-ICA values stayed above the control at every well-covered electrode (panels b, d), so an EEG-ECG relationship persists after cleaning, in agreement with Section 3.3 and Figure S2. Mutual information, which also captures nonlinear dependence, followed the same order as correlation, giving no sign of dependence beyond the linear measures.

S4. Power spectral density: pre-ICA, post-ICA, non-locked control, and ECG

Welch power spectral density was computed for the pre-ICA, post-ICA, and non-locked evoked waveforms of Section S3 and for the ECG evoked average, in the same 150-patient subsample.

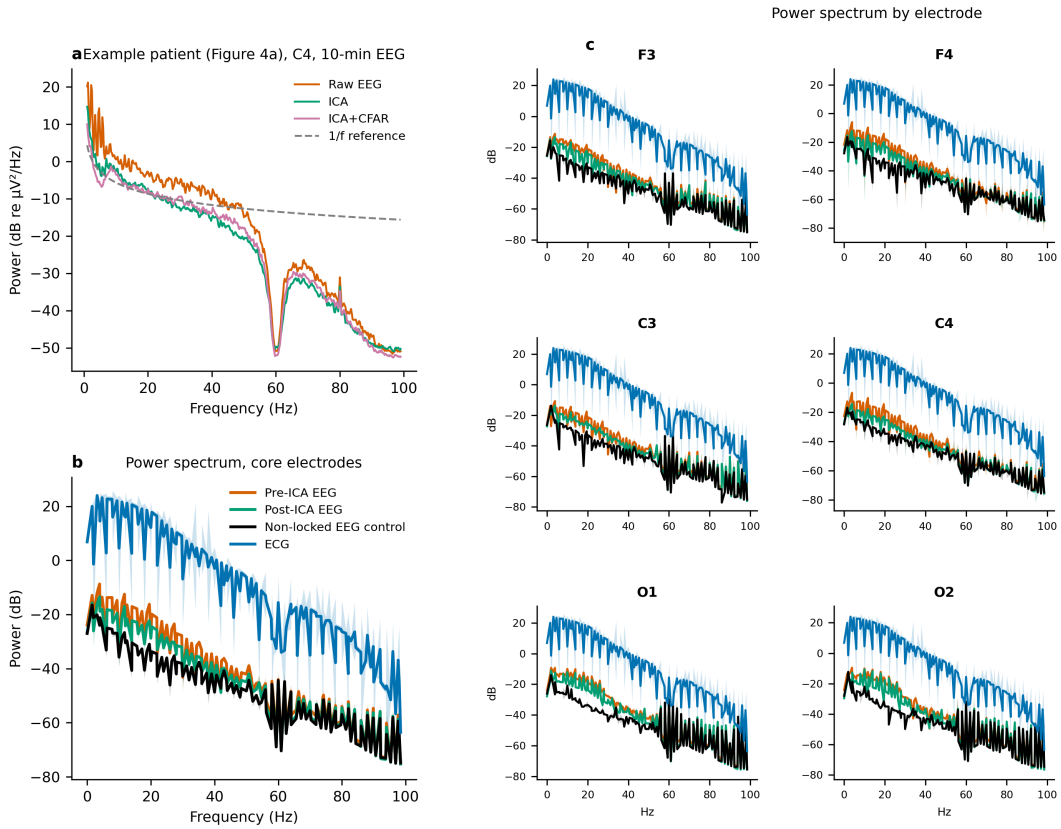


Figure S4. Power spectral density (Welch, dB). (a) Continuous 10-min EEG of the Figure 4a example patient (C4): raw, after ECG-informed ICA, and after ICA+CFAR; dashed line, 1/f spectrum fit to ICA+CFAR over 1–40 Hz. (b, c) Pre-ICA, post-ICA, non-heartbeat-locked control, and ECG evoked waveforms (n = 150): (b) core electrodes (F3/F4/C3/C4), mean ± SEM; (c) each of the six well-covered sites.

The ECG spectrum was broadband, dominated by the sharp QRS transient, which the non-locked control lacked. Pre-ICA and post-ICA EEG spectra lay between these references and closer to the ECG than the control, supporting Sections 3.3 and S3 in the spectral domain. Spectral shape was similar across electrodes (panel c), so CFA varies across the scalp mainly in magnitude (Figure 2c), not spectral content. Spectra are shown up to 99 Hz, just below the 100 Hz upper edge of the analysis filter (Section 2.2).

S5. Adjustment of the sex effect for BMI

Men had higher CFA R^2 than women (Section 3.4), but women had a higher mean BMI (34.8 vs. 31.8 kg/m²), and BMI itself was associated with CFA. In patients with BMI (n = 6,127), OLS regression of patient-mean CFA R^2 on sex gave a male-versus-female coefficient of 0.031 (95% CI 0.020 to 0.041, p = 1.1e-08). Adding BMI did not attenuate it (0.040; 95% CI 0.029 to 0.050, p = 3.2e-13), and BMI was an independent predictor (0.0029 per kg/m², p = 1.5e-21).

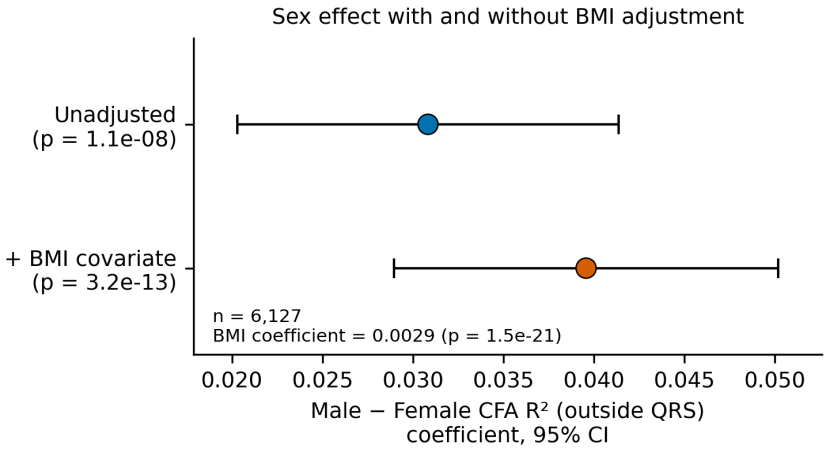


Figure S5. Male-versus-female CFA R² coefficient without and with BMI adjustment (± 95% CI).

The sex coefficient rose slightly after BMI adjustment, as expected given women's higher BMI, so BMI does not explain the sex effect; sex and BMI were independently associated with CFA.

S6. How CFAR works

CFAR (Section 2.8) is a single linear layer whose weights are learned from ECG-locked EEG. Training passes each patient's continuous EEG through R-peak epoching, averaging, and an SVD that returns the patient's CFA scalp pattern; patterns are sign-aligned and averaged with CFA R² weights into the population filter u. Inference needs no ECG: every EEG sample x is replaced by (I − u u■) x. The filter converged quickly: with 100 training patients its median cosine similarity to held-out patients' own patterns was 0.93, against 0.93 with the full training set (n = 6,711 patterns). Taking CFA as detected when a patient's R² exceeded the 95th percentile of the non-heartbeat-locked null ($R^2 > 0.23$), CFA was detected in 64% of patients in raw EEG and 29% after CFAR; CFAR cleared it in 56% of patients in whom it was detected (2,409 of 4,278), and 102 patients without detected CFA rose above the threshold after cleaning.

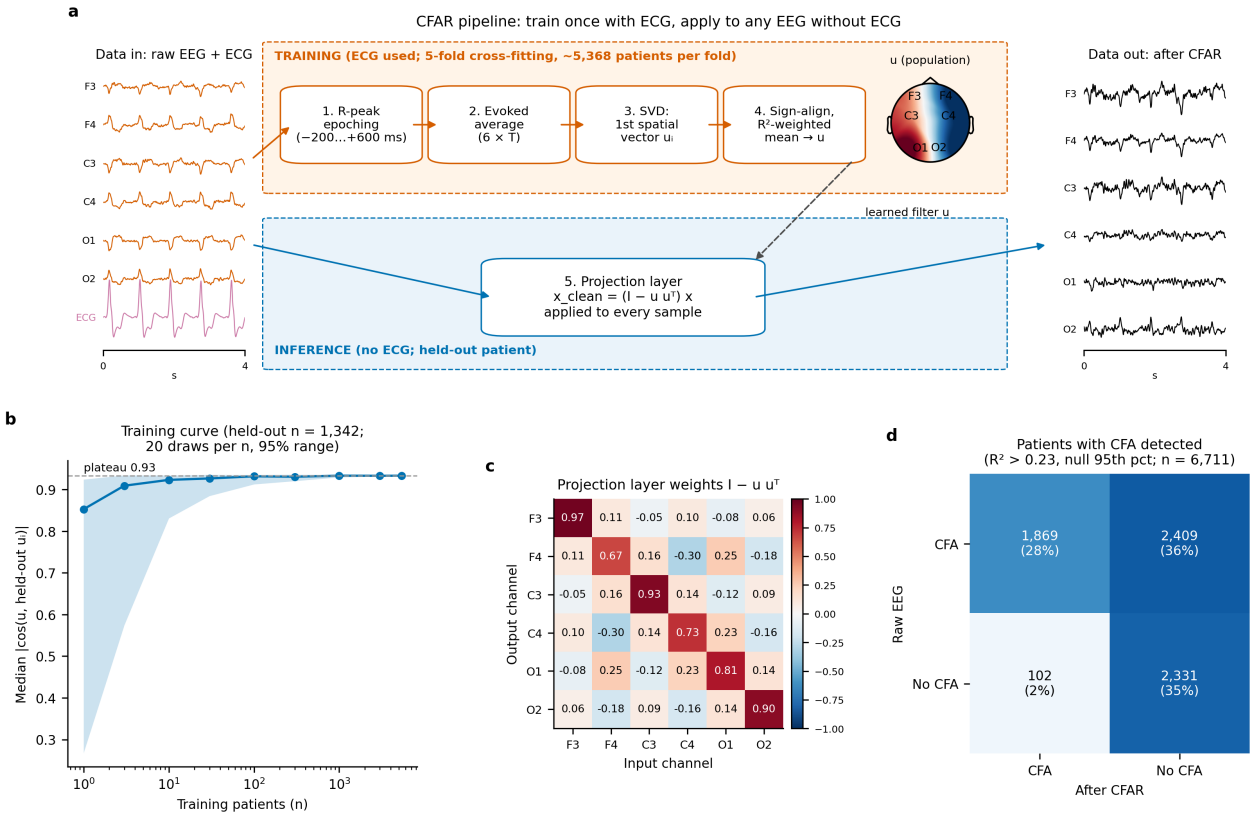


Figure S6. How CFAR works. (a) Pipeline, with 4 s of the Figure 4a example patient: data in (six-channel EEG and ECG), training layers (orange; ECG used, 5-fold cross-fitting), the learned population filter u, and the inference layer (blue; no ECG) producing cleaned EEG. (b) Training curve: median |cosine| between the filter learned from n training patients and held-out patients' own CFA patterns (20 random draws per n; band, 95% range). (c) Weights of the projection layer I – u u■. (d) Confusion matrix of patients with CFA detected (R^2 above the 95th percentile of the non-heartbeat-locked null) in raw EEG versus after CFAR (n = 6,711).

S7. CFA R^2 across segment durations

Extending Figure 3d, the full-cohort analysis was repeated at 5, 10, 20, 30, 45, 60, 75 and 90 min. Each duration drew its own seeded random window, so duration and position in the night are not separated here. To keep differences in channel sets from appearing as duration effects, only patient-channel pairs present at every duration were used (n = 6,768 patients). Mean R^2 rose from 0.31 at 5 min to 0.44 at 90 min, with smaller gains at longer durations. Rank agreement with the 90-min estimate rose from ρ = 0.78 (5 min) to 0.98 (75 min), while the share of patients with a valid window fell from 100% to 94%. Because agreement with the longest duration always favours the duration closest to it, this comparison cannot by itself identify an optimal duration; Section S8 tests this directly.

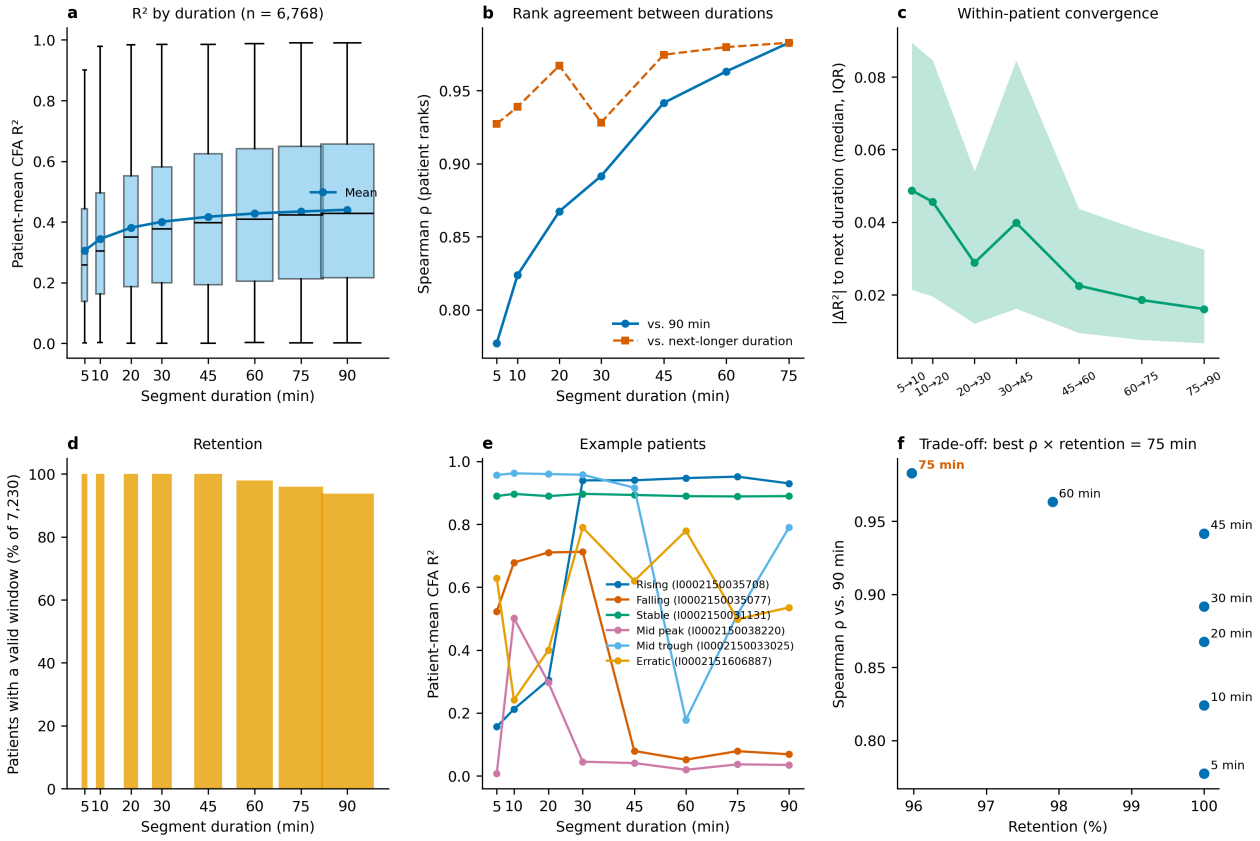


Figure S7. CFA R² across segment durations (n = 6,768 patients with every duration). (a) Patient-mean R² per duration (box, IQR; line, mean). (b) Spearman rank agreement with the 90-min estimate and with the next-longer duration. (c) Per-patient |ΔR²| between consecutive durations (median, IQR band). (d) Share of the 7,230-patient list with a QC-passing window. (e) Example patients whose R² changes differently with duration. (f) Rank agreement with the 90-min estimate vs. retention.

S8. Test-retest reliability and the optimal window length

To separate duration from position and to measure reliability, each recording was divided into 5-min blocks, and each block was checked with the same quality criteria as the main analysis (Section 2.2). For each block, the R-peak-locked EEG and ECG epochs were summed. A window of L minutes was then defined as L/5 contiguous QC-passing blocks, and its evoked average was the pooled sum of those blocks divided by their total beat count. All durations were therefore measured within the same night, with nested windows and as many non-overlapping repeats as the recording allowed. CFA R² (outside QRS, mean across channels) was computed as in Section 2.3. Reliability was the one-way random-effects ICC(1) across a patient's repeat windows (patients with at least two windows; 95% CI from 200 patient bootstraps).

ICC rose steadily with duration, from 0.70 (95% CI 0.69–0.70) at 5 min to 0.74 at 10 min and 0.86 (0.85–0.87) at 90 min. No single duration reached 0.90. Window-to-window SD within patients fell only slightly (0.122 to 0.098), whereas between-patient SD grew (0.184 to 0.242). In the same patients and night (n = 6,175), mean R² still rose with duration (panel c), so the duration effect in Figure 3d is at least partly an estimation effect: averaging more beats reduces residual EEG noise in the evoked average and raises its correlation with the ECG. Longer windows also cost patients: 85% had a QC-passing 90-min stretch, against 99% at 30 min. The shortest duration whose ICC lower bound reached 0.75 was 15 min (ICC 0.77, 99.8% retained). The Spearman-Brown formula predicts that averaging separate 5-min windows spread across the night would reach ICC 0.90 with 20 min of data in total, which no single contiguous window achieved. This is a prediction that assumes independent windows and was not tested empirically.

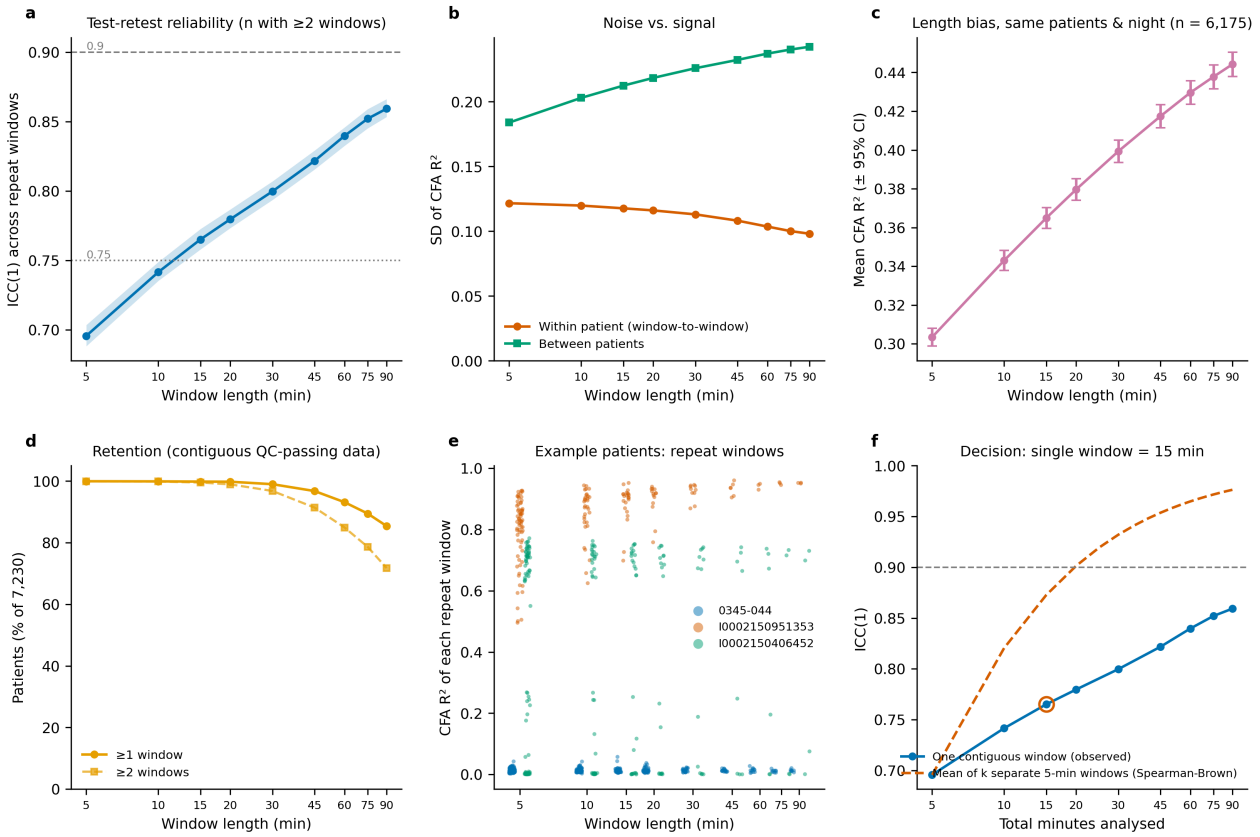


Figure S8. Test-retest reliability of CFA R² vs. window length (n = 7,226 patients; repeat windows from the same night). (a) ICC(1) across repeat windows (band, 95% bootstrap CI; dashed/dotted, 0.90/0.75). (b) Within-patient (window-to-window) and between-patient SD. (c) Mean R² in patients with a window at every duration (± 95% CI). (d) Patients with at least one / two windows of each duration. (e) Every repeat window of three example patients (low, median, and high 5-min spread). (f) ICC of one contiguous window vs. the Spearman-Brown prediction for the mean of k separate 5-min windows, against total minutes analysed; circle, the shortest single window reaching ICC lower bound 0.75.

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References

1. Dirlich G, Vogl L, Plaschke M, Strian F. Cardiac field effects on the EEG. Electroencephalogr Clin Neurophysiol. 1997;102(4):307-315.

2. Dirlich G, Dietl T, Vogl L, Strian F. Topography and morphology of heart action-related EEG potentials. Electroencephalogr Clin Neurophysiol. 1998;108(3):299-305.

3. Urigüen JA, Garcia-Zapirain B. EEG artifact removal—state-of-the-art and guidelines. J Neural Eng. 2015;12(3):031001. doi:10.1088/1741-2560/12/3/031001.

4. Jiang X, Bian G-B, Tian Z. Removal of artifacts from EEG signals: a review. Sensors. 2019;19(5):987. doi:10.3390/s19050987.

5. Schandry R, Sparrer B, Weitkunat R. From the heart to the brain: a study of heartbeat contingent scalp potentials. Int J Neurosci. 1986;30:261-275.

6. Park H-D, Blanke O. Heartbeat-evoked cortical responses: underlying mechanisms, functional roles, and methodological considerations. NeuroImage. 2019;197:502-511.

7. Coll M-P, Hobson H, Bird G, Murphy J. Systematic review and meta-analysis of the relationship between the heartbeat-evoked potential and interoception. Neurosci Biobehav Rev. 2021;122:190-200.

8. Steinfath TP, et al. Heartbeat-evoked responses in M/EEG: a systematic review of methods with suggestions for analysis and reporting. Psychophysiology. 2026;63(4):e70297. doi:10.1111/psyp.70297.

9. Virjee R-I, Kandasamy R, Garfinkel SN, Carmichael DW, Yogarajah M. Review of methods to derive the heartbeat-evoked potential: past practices and future directions. Soc Cogn Affect Neurosci. 2026:nsag057. doi:10.1093/scan/nsag057.

10. Kern M, Aertsen A, Schulze-Bonhage A, Ball T. Heart cycle-related effects on event-related potentials, spectral power changes, and connectivity patterns in the human ECoG. NeuroImage. 2013;81:178-190.

11. Buot A, Azzalini D, Chaumon M, Tallon-Baudry C. Does stroke volume influence heartbeat evoked responses? Biol Psychol. 2021;165:108165. doi:10.1016/j.biopsycho.2021.108165.

12. Tochikubo O, Miyajima E, Shigemasa T, Ishii M. Relation between body fat-corrected ECG voltage and ambulatory blood pressure in patients with essential hypertension. Hypertension. 1999;33(5):1159-1163. doi:10.1161/01.HYP.33.5.1159.

13. Fraley MA, Birchem JA, Senkottaiyan N, Alpert MA. Obesity and the electrocardiogram. Obes Rev. 2005;6(4):275-281.

14. Boudewyn MA, Luck SJ, Farrens JL, Kappenman ES. How many trials does it take to get a significant ERP effect? It depends. Psychophysiology. 2018;55(6):e13049.

15. Li Q, Wen S, Sun H, Ganglberger W, Tripathi A, Turley N, et al.; Westover MB. The Human Sleep Project (HSP). Brain Data Science Platform. 2026. doi:10.60508/m3sw-rz13.

16. Gramfort A, et al. MEG and EEG data analysis with MNE-Python. Front Neurosci. 2013;7:267.

17. Hyvarinen A, Oja E. Independent component analysis: algorithms and applications. Neural Netw. 2000;13(4-5):411-430.

18. Ablin P, Cardoso J-F, Gramfort A. Faster independent component analysis by preconditioning with Hessian approximations. IEEE Trans Signal Process. 2018;66(15):4040-4049.

19. Kraskov A, Stögbauer H, Grassberger P. Estimating mutual information. Phys Rev E. 2004;69(6):066138.

20. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Series B. 1995;57(1):289-300.

21. Vallat R, Walker MP. An open-source, high-performance tool for automated sleep staging. eLife. 2021;10:e70092. doi:10.7554/eLife.70092.

22. Pérez JJ, Guijarro E, Barcia JA. Suppression of the cardiac electric field artifact from the heart action evoked potential. Med Biol Eng Comput. 2005;43(5):572-581.

23. Arnau S, Sharifian F, Wascher E, Larra MF. Removing the cardiac field artefact from the EEG using neural network regression. Psychophysiology. 2023:e14323. doi:10.1111/psyp.14323.

24. Gabriel S, Lau RW, Gabriel C. The dielectric properties of biological tissues: II. Measurements in the frequency range 10 Hz to 20 GHz. Phys Med Biol. 1996;41(11):2251-2269.